

Quiz 6 Reflection

SCIE 001 Physics

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1. Three Oscillators

1.1. Specify Question

Question 1: Three Oscillators

Consider 3 identical masses attached to 3 identical springs, each with identical speed at the moment shown below. The dashed line shows the equilibrium position and there is no damping. What can you say about the **maximum velocity** attained by each of these oscillators?

☐ $v_3 > v_1 > v_2$

☐ $v_1 > v_2 = v_3$

☐ $v_2 > v_1 > v_3$

☐ $v_2 > v_3 > v_1$

☐ $v_2 = v_3 > v_1$

☐ Not enough information to answer.

☐ $v_1 = v_2 = v_3$

☒ $v_3 > v_2 > v_1$ ✖

✖ 0%

Correct answer

$v_2 = v_3 > v_1$

1.2. Diagnosis Phase

I misread the diagram and incorrectly thought that the 3rd mass-spring system was stretched further than the second spring at the time of the snapshot.

Maximum velocity is directly proportional to maximum amplitude.

$$v_{\text{max}} = A\omega$$

In such a scenario, the third spring would reach a greater amplitude than the first spring as, since the velocity

Words: 53

1.3. Correction Phase

I was incorrect in reading that the

Words: 7